



MONITOR

A well-configured monitor will give you clear and sharp images—otherwise, it's bleary eyes and headaches for you

Your window into a world of movies, graphics and images on your computer, the monitor is one of the least regarded devices of your system, considering the fact it is your only portal to the outcome of that sea of digital information in the beige box. Therefore, the performance of your monitor can play a subtle, yet significant role in how efficiently you can work in the long run.

Have you ever felt the need to have more information packed onto your screen so that you don't have to scroll so much through your spreadsheet? The factor that can facilitate this is the resolution of your monitor. The resolution is simply the number of horizontal and vertical pixels that can be drawn on your screen—the higher this number, the greater is the amount of information you can perceive. Therefore, simply increasing your screen resolution will give you the ability to view more in applications such as spreadsheets, word processors and even Web pages. However, increasing resolution does come with a trade-off—a decrease in refresh rate.

If your monitor is operating at anything less than 75 Hz, the human eye will be able to perceive a flicker in the display. While this may not be apparent at first, staring at such a display for hours on end can cause serious eye fatigue. The

specifications of all monitors state the maximum resolution supported at a given refresh rate, because they have a fixed video bandwidth—an increase in display resolution will result in a corresponding reduction in the refresh rate

since the video bandwidth always remains constant. Therefore, if you run your monitor at its highest possible resolution, you will be also running it at its lowest refresh rate, which could cause eyestrain.

Even though monitors have a significantly higher resolution as compared to television, nothing beats watching a movie on a larger screen. With the resurgence in newer forms of multimedia, life-like gaming and the pro-

liferation of DivX and DVD movies on PC-based home theatres, the days of the 15-inch monitor are gone. While it is not a direct bottleneck (Microsoft Word will still take the same amount of time to start!) having a larger screen is definitely a big advantage.

Over time, a monitor's display gets slightly distorted due to ageing of the picture tube and other electronic components. Therefore, you need to have an adequate number of controls that allow the different aspects of the display to be controlled. Besides the horizontal and vertical size and position, there are numerous other controls that can manipulate how the display looks and fits into the frame of the screen. This would include parameters such as raster rotation, trapezoidal correction, parallelogram correction, pincushioning and pin-balancing. The presence of gamma correction and colour temperature controls also ensures that the display is as lifelike and balanced as possible.

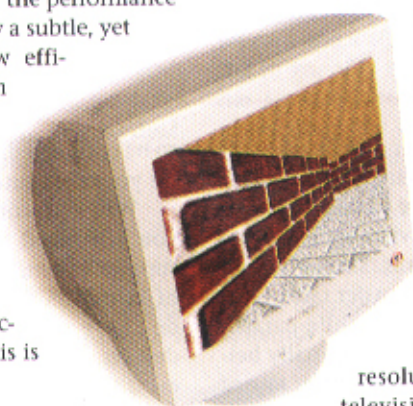
The renewed interest in TFT monitors

has also raised questions with regards to which of the two technologies is better. While there is no definite answer, each technology exhibits specific advantages in certain areas. While CRTs offer the advantage of smaller dot-pitches and higher resolutions, TFTs exhibit inherently better brightness and contrast levels allowing for much more vivid images. Also, TFTs are lighter and have form factors and footprints that are significantly smaller than in their CRT counterparts. On the other hand, one of the two most prominent disadvantages with TFTs is the display 'response time'—the time taken for the TFT pixel elements to switch on or off. This results in varying levels of ghosting in TFTs, especially where there are fast-moving images as in movies and gaming. TFTs are also much more expensive compared to CRTs. However, TFT prices have been on the decline and a time is predicted where LCD technology will become as economically viable as CRT display systems.

Tips & Tweaks

- Whatever resolution you use, your monitor's refresh rate should be at least 75 Hz. Anything less could result in eyestrain after prolonged use.
- The lighting around your workspace should not be significantly brighter or dimmer than the brightness of your monitor. This helps minimise eyestrain.
- Keep your monitor away from electrical components such as transformers and switching devices as they affect the display and could even be detrimental to the picture tube in the long run. Degaussing it occasionally can reduce the colour patches but refrain from doing this too frequently.
- Since the monitor is probably the last component you would ever upgrade, it makes sense to buy the largest monitor you can afford when buying a new computer.

The larger the picture tube, the greater is the refresh rate and resolution



TFT monitors offer greater brightness and contrast levels as compared to CRT monitors

